FINTRUST PAYMENT SOLUTIONS

Regulatory Briefing: Fraud Alert Explanation

Transactions TXN_007815 and TXN_003299

Prepared for James Carson, Federal Trade Commission

Course project — fictional scenario

Briefing Scope and Agenda

This briefing addresses the two specific fraud alerts requested by the FTC for review.

1	**System Methodology**	Isolation Forest algorithm and SHAP explainability framework
2	**TXN_007815 — Detailed Findings**	Contributing factors, SHAP values, and transaction data
3	**TXN_003299 — Detailed Findings**	Contributing factors, SHAP values, and transaction data
4	**Fairness & Non-Discrimination**	FCRA/ECOA compliance and protected characteristic exclusions
5	**Human Oversight Process**	Analyst review workflow and escalation procedures
6	**Model Limitations**	What the model cannot determine and acknowledged constraints

System Methodology

Detection: Isolation Forest

Algorithm Type:
Unsupervised anomaly detection

Mechanism:
Isolates transactions by randomly selecting features and split values. Anomalous transactions require fewer splits to isolate.

Configuration:

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200 estimators
20 behavioral features
10% contamination rate
Trained on 10,000 transactions
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Explainability: SHAP

Framework:
SHapley Additive exPlanations (Lundberg & Lee, 2017)

Foundation:
Based on Shapley values from cooperative game theory. Provides mathematically rigorous, additive feature attribution.

Output:
Per-feature contribution to each individual fraud decision. Positive SHAP = increases fraud risk. Negative SHAP = decreases fraud risk.

Transaction TXN_007815 — Ove

FLAGGED — HIGH RISK

Field	Value	Context
Transaction ID	TXN_007815	Unique identifier
Amount	$279.88	Above avg ($151.26 for legit)
Time	22:00 (10 PM)	Late evening transaction
Day of Week	Day 3 (Thursday)	Weekday
Distance from Home	86.3 km	Avg legit: 15.1 km
International	Yes	Cross-border transaction
Device Change	Yes	Unrecognized device used
Merchant Category	Entertainment	—
Account Age	1,974 days (5.4 yrs)	Established account
Anomaly Score	-0.1201	Below threshold (flagged)
Model Decision	Fraud	Confirmed by investigation

TXN_007815 — SHAP Contributing Factors

Baseline E[f(X)] = -11.373 | Final f(x) = -6.858 | Net fraud risk contribution: +4.515

Rank	Feature	Value	SHAP	Direction
1	distance_from_last_transaction_km	11.423 (scaled)	+1.29	**INCREASES risk**
2	device_change	3.937 (= Yes)	+1.22	**INCREASES risk**
3	merchant_entertainment	3.27 (= Yes)	+1.03	**INCREASES risk**
4	is_international	3.205 (= Yes)	+0.95	**INCREASES risk**
5	distance_from_home_km	3.709 (86.3 km)	+0.79	**INCREASES risk**
6	hour_of_day	1.582 (22:00)	+0.28	**INCREASES risk**
7	transaction_amount	1.23 ($279.88)	+0.08	**INCREASES risk**
8	merchant_grocery	-0.478	-0.19	Decreases risk
9	merchant_online	-0.398	-0.17	Decreases risk
10	merchant_retail	-0.431	-0.16	Decreases risk

Top 5 factors all increase fraud risk. Geographic distance and device change are the primary drivers.

Transaction TXN_003299 — Ov

FLAGGED — HIGH RISK

Field	Value	Context
Transaction ID	TXN_003299	Unique identifier
Amount	$400.44	2.6x avg legit ($151.26)
Time	10:00 (10 AM)	Business hours
Day of Week	—	—
Distance from Home	122.5 km	Avg legit: 15.1 km (8x above)
International	Yes (scaled: 3.205)	Cross-border transaction
Device Change	No	Recognized device
Merchant Category	Restaurant	—
Account Age	658 days (1.8 yrs)	Moderately established
Transactions (24h)	7	Avg legit: 2.5 (2.8x above)
Anomaly Score	-0.0918	Below threshold (flagged)
Model Decision	Fraud	Confirmed by investigation

TXN_003299 — SHAP Contributing Factors

Baseline E[f(X)] = -11.373 | Final f(x) = -7.539 | Net fraud risk contribution: +3.834

Rank	Feature	Value	SHAP	Direction
1	is_international	3.205 (= Yes)	+1.25	**INCREASES risk**
2	distance_from_home_km	5.644 (122.5 km)	+1.25	**INCREASES risk**
3	merchant_restaurant	2.219 (= Yes)	+0.79	**INCREASES risk**
4	transactions_last_24h	2.751 (7 txns)	+0.58	**INCREASES risk**
5	transaction_amount	2.443 ($400.44)	+0.50	**INCREASES risk**
6	transactions_last_7d	2.475 (scaled)	+0.46	**INCREASES risk**
7	distance_from_last_transaction_km	1.437 (scaled)	+0.19	**INCREASES risk**
8	merchant_grocery	-0.478	-0.24	Decreases risk
9	merchant_online	-0.398	-0.19	Decreases risk
10	merchant_retail	-0.431	-0.17	Decreases risk

Multi-factor pattern: international + geographic anomaly + high velocity + elevated amount.

FCRA / ECOA Compliance Framework

Transparency

SHAP values provide feature-by-feature breakdown for every decision. Waterfall plots document exact contribution of each factor. No black-box elements.

Fairness & Non-Discrimination

Model uses only behavioral/transactional features. No protected characteristics (race, gender, age, ethnicity, religion) are used as inputs. Uniform evaluation criteria applied to all cardholders.

Consistency & Reproducibility

Shapley values ensure stable, mathematically consistent attributions. All parameters documented and version-controlled (v2.1). Results reproducible for independent audit.

Cardholder Rights

Right to explanation of specific factors. Right to dispute within 30 days. Right to human review. Right to documentation. Fraud alerts do not impact credit scores.

Human Oversight Process

Automated detection is the first step — human review is the final authority.

Model Scores Transaction

20 features evaluated

→

SHAP Generates Explanation

Per-feature attributions

→

Analyst Reviews High-Risk Cases

Trained fraud specialists

→

Final Decision & Documentation

Audit trail preserved

Key Safeguards

- All high-risk flagged transactions undergo certified fraud analyst review before account action
- Borderline cases (near decision threshold) require mandatory human judgment — the model recommends, humans decide
- Complete audit trail: model score, SHAP explanation, analyst decision, and rationale are logged for every case
- Cardholders may request human re-review within 30 days of any fraud alert

Model Limitations and Constraints

FinTrust acknowledges the following limitations in its fraud detection system:

Cannot Determine Intent

The model identifies anomalous patterns but cannot determine whether a transaction is actually fraudulent. It flags statistical outliers — the distinction between fraud and unusual-but-legitimate behavior requires human judgment.

Threshold Sensitivity

Transactions near the decision boundary (anomaly score near 0) may be classified differently with minor parameter changes. Borderline cases are explicitly routed to human review to mitigate this.

Training Data Scope

Model was trained on 10,000 FinTrust transactions. Novel fraud patterns not represented in this dataset may not be detected until the model is retrained (quarterly schedule).

Feature Limitations

The system evaluates 20 predefined features. Fraud tactics that exploit dimensions not captured by these features may evade detection. FinTrust continuously evaluates new feature candidates.

Correlation vs. Causation

SHAP values indicate statistical associations, not causal relationships. A high SHAP contribution means the feature correlates with anomalous behavior, not that it caused fraud.

Summary and Availability

Both transactions (TXN_007815 and TXN_003299) were flagged through a consistent, documented, and auditable process.

Each decision includes specific contributing factors with quantified weights, compliant with FCRA adverse action notice requirements.

Human oversight ensures no automated decision is final without analyst review.

FinTrust is prepared to provide:

Full SHAP output files • Model documentation • Audit trail records • Analyst review logs

FinTrust Compliance Office: compliance@fintrust.example.com